



Trimble Placement Experience

- Varun Karthik T (Batch 2022- 26)

About the Company

Trimble is a global technology company specializing in positioning, navigation, and mapping solutions. Founded in 1978, it was among the first to commercialize GPS technology for civilian use. The company integrates various technologies like GPS, laser, optical, and inertial systems with software and services to enhance productivity across different sectors.

Trimble's solutions are widely used in industries such as construction, agriculture, geospatial mapping, and transportation. A core focus of the company is connecting the physical and digital worlds to improve efficiency and decision-making.

Previous Experience : I had previously interviewed at SAP Labs, where I was asked a dynamic programming problem. While I couldn't provide the optimal solution, I successfully answered all the other questions

Trimble Recruitment Process

The recruitment process for the **Software Engineer** role consisted of one coding round followed by three interview rounds.

Round 1: Coding Round

The coding round was conducted on the Codility platform, which featured a built-in AI assistant. There were two coding questions.

- **Question 1:** Given a string containing '?' and an integer 'k,' you can replace each '?' with any alphabet. Additionally, at most 'k' other characters in the string could be replaced to make it a palindrome. The task was to return the resulting palindrome if possible; otherwise, return 'NO'
- **Question 2:** Given an $m \times n$ matrix, return the count of possible sub-matrices where the sum of the elements inside the sub-matrix is equal to the sum of the elements outside of it.

I successfully solved both questions and submitted my solutions two minutes before the deadline. The built-in AI assistant on Codility helped me optimize my approach for the second problem.

Round 2: Technical Interview (60 minutes)

35 people were shortlisted for this round. It was a 1:1, in-person interview held at CUIC. The interviewer was very calm and gave me time to think, even for simple questions. After a discussion about my coding round experience, we moved directly to the technical questions. He asked me to write out and explain most of my solutions.

The questions covered the following areas:

- **Data Structures & Algorithms (DSA):** Luckily, He asked about the different approaches I knew for calculating a Fibonacci number. I explained the recursive, dynamic programming, and iterative methods. He then asked me to write all three approaches and discuss their respective time and space complexities.
- **Sorting:** The interviewer asked about my favorite sorting algorithm. I chose Merge Sort, and he then provided an array of about 15 elements, asking me to perform a step-by-step walkthrough of the sorting process, including a complete simulation.
- **Logical Reasoning:** I was given a tank with an unlimited supply of water, a 5L cup, and a 3L cup. The task was to measure exactly 4L of water.
- **Projects:** He asked me to draw a high-level system design for one of my projects and explain the data flow for a specific feature.
- **Object-Oriented Programming (OOPS):** When I mentioned that OOPS was my favorite subject, he asked the following questions:
 - What is the difference between classes and objects?
 - What is a singleton class and where is it used?
 - What are abstract methods?
- **Operating Systems (OS):**
 - What is a race condition?
 - What is the difference between concurrency and parallelism?
- **Database Management Systems (DBMS):**
 - What types of databases have you worked with?
 - What is the difference between PostgreSQL and MongoDB?
 - Why do we need normalization? What is denormalization?
 - Why did you choose a particular database for your project?
 - What is the purpose of Object-Relational Mapping (ORM)?

I was able to answer most of the questions correctly. The questions in this round were not overly difficult and covered the fundamentals of all core computer science areas. The round ended with me asking the interviewer some questions.

Round 3: System Design and Managerial Round (40-45 minutes)

Around 20 candidates were shortlisted for this round. This round was enjoyable, and the interviewer was very friendly; it felt more like a casual chat than a formal assessment. He recognized me from the pre-placement talk because I had asked him a few questions then.

- **Projects & Hobbies:** We started with an introduction where we discussed my hobby, photography. He then asked me to explain the same project discussed in the previous round and inquired about the different approaches my team and I considered for a particular feature.
 - **System Design:** I was asked to create a system design for a mall parking system.
 - **Managerial & Scenario-Based Questions:**
 - "If your full-time offer depended on your internship performance, how would you approach your work during the internship?"
 - "How do you go about learning a new skill?"
 - "Imagine you are a Trimble employee with 10 years of experience. A new problem is presented to all employees, and a fresher's idea is better than yours. What would you do?"
 - A follow-up question was: "How do you determine if an idea is good?"
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Round 4: HR - Personal Experience Round (30 minutes)

11-13 people were shortlisted for this final round. The discussion was centered around my personal experiences and background.

- She asked about my hackathon experiences, my weakness, and my family background
- She asked why I chose to study at CEG
- She asked a hypothetical question: "If you received another offer, which company would you choose and why?"

- The HR representative noticed that my friend Ajay, who is also my hackathon teammate, and I had worked on common projects and asked us about our shared experiences. (We both ultimately received an offer) .
 - We also discussed my preferred domains of work.
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In the end, **six of us received an offer**, and I am happy to be one of them. The interview process primarily focused on my understanding of core concepts, system design capabilities, and the projects I had worked on.

My Preparation Strategy:

- **DSA:** I followed Striver A2Z sheet and NeetCode 150 🙏
- **Core Concepts:** I relied on GeeksforGeeks and my academic coursework.
- **SQL:** I solved some questions on the "SQL 50" list on LeetCode
- **Projects:** My deployed projects and hackathon experience were incredibly helpful during the interviews.

Tips:

- Don't over-exaggerate your resume. Even if it's minimal, make sure you truly understand everything you put on it.
- Don't get demotivated if you don't get placed early; stay confident and focus on the process.
- Don't get distracted by the number of problems solved. Some people find patterns by solving just a few problems, while others need to solve many more. Understand who you are and put in effort accordingly.
- You cannot prepare for everything in just one month before placements; how you spend your entire time in college will help you in various ways.
- Meaningful experiences from projects, hackathons, and clubs cannot be gained at the last minute. My participation in hackathons, in particular, taught me a lot.

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